EQUIVARIANT ISOMETRIC EMBEDDINGS VIA GÜNTHER'S METHOD

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Abstract. Moore and Schlafly proved in 1980 that a compact Riemannian manifold on which a compact Lie group G acts by isometries has an isometric embedding into a Euclidean space that is equivariant with respect to an orthogonal representation of G. The original proof used the Nash–Moser implicit function theorem with smoothing operators. We give a proof that replaces Nash–Moser by Günther's method, which reduces the isometric embedding equation to a fixed point problem for a contraction on a Banach space of Hölder maps. All of Günther's operators commute with the group action, so the contraction preserves the closed subspace of equivariant maps, and the uniqueness of the small fixed point makes the correction equivariant without any averaging of nonlinear solutions. We also formalize the algebraic and functional-analytic core of the argument in Lean 4 with Mathlib: the equivariant Günther fixed point theorem, the equivariance and coordinate independence of the Nash–Günther pointwise right inverse, the freeness of the Veronese map, the Haar-averaged inner product used in the approximation step, and the assembly of the main theorem from explicitly stated analytic inputs. The formal proofs use only the standard axioms of Lean.

1. Introduction

The Nash embedding theorem [12] says that every Riemannian manifold has an isometric embedding into a Euclidean space of sufficiently large dimension. In [10] J. D. Moore and the author proved an equivariant version, extending the case of homogeneous spaces treated by Moore [9].

Main Theorem (Moore–Schlafly [10])**.** *Let (M, g) be a compact Riemannian manifold and let G be a compact Lie group acting smoothly on M by isometries. Then there are an orthogonal representation $\rho\colon G \to O(N)$ and a smooth embedding $X\colon M \to \mathbb{R}^N$ such that*

$$X^*\langle\cdot,\cdot\rangle_{\mathbb{R}^N} = g \qquad \text{and} \qquad X(\sigma p) = \rho(\sigma)\, X(p) \quad \text{for all } \sigma \in G,\ p \in M.$$

The proof in [10] has two steps. Step 1 finds a "perturbable" (free) equivariant embedding X_0 and shows, using the Nash–Moser implicit function theorem in the form of Sergeraert [16], that every invariant metric close to $F(X_0)$ is induced by an equivariant embedding. Smoothing operators are made equivariant by composing with the Haar projection. Step 2 shows that invariant metrics induced by equivariant maps are dense, using the spectral theory of the Laplacian and Haar averaging.

In 1989 Günther [4, 5] found a much simpler proof of the Nash embedding theorem. He rewrote the linearized problem so that the nonlinear equation becomes a fixed point problem

$$v = \mathcal{M}(h) + \mathcal{B}(v, v) \tag{1}$$

in which $\mathcal{B}$ loses no derivatives. The ordinary contraction mapping principle on a Hölder space then replaces the Nash–Moser iteration. Expositions are given by Yang [18] and Tao [17]; both

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note that Günther's reduction to a torus is a convenience and that one can work directly on M with a background metric.

This paper does two things.

(1) *A proof of the Main Theorem by Günther's method* (Sections 2–8). We work intrinsically on M, using the Levi-Civita connection of the G-invariant metric g and the operator $(\Delta-1)^{-1}$. Every operator in Günther's construction is then natural under isometries, so the map $v\mapsto \mathcal{M}(h)+\mathcal{B}(v,v)$ commutes with the action of G on maps $M\to\mathbb{R}^N$. Its unique small fixed point is therefore equivariant (Theorem 6.1). No smoothing operators are needed and no nonlinear solution is averaged. Step 2 is the same as in [10].

(2) *A machine-checked formalization* of the algebraic and functional-analytic core of this proof in Lean 4 [7] with Mathlib [8] (Section 9). Mathlib does not yet contain Hölder spaces of sections over manifolds, the Laplace–Beltrami operator, Schauder estimates or the Mostow–Palais theorem. We therefore prove the new equivariant ingredients unconditionally, and we prove the Main Theorem conditionally: the classical analytic inputs are explicit hypotheses, and none of them asserts the existence of an equivariant solution of the nonlinear problem.

Equivariant versions of Günther's construction for cocompact discrete group actions on noncompact manifolds have been studied recently by Qiu [14]; see also Burago–Qiu [2]. As in [10], no bound on N depending only on $\dim M$ is possible in the equivariant setting: [10, §6] constructs left-invariant metrics on S^3 that need arbitrarily large N.

Conventions. All manifolds are smooth, compact and without boundary (see Remark 8.1 for boundary), and $n=\dim M$. Repeated indices are summed and indices are raised with g. "Smooth" means C^∞. We fix $\alpha\in(0,1)$.

2. Setup

2.1. **Equivariant maps and induced metrics.** Let (V,ρ) be a finite-dimensional orthogonal representation of G. A map $X\colon M\to V$ is *equivariant* if $X(\sigma p)=\rho(\sigma)X(p)$. Its *induced metric* is the symmetric 2-tensor

$$F(X)=\langle dX,dX\rangle,\qquad F(X)_{ij}=\langle \partial_i X,\partial_j X\rangle.$$

If X is equivariant then $F(X)$ is G-invariant, because $\rho(\sigma)$ is orthogonal. Two maps $X_1\colon M\to V_1$ and $X_2\colon M\to V_2$ give $(X_1,X_2)\colon M\to V_1\oplus V_2$ with

$$F(X_1,X_2)=F(X_1)+F(X_2),\qquad F(tX)=t^2F(X). \tag{2}$$

If X_1 is an embedding, so is (X_1,X_2). A G-invariant symmetric 2-tensor is *realizable* if it equals $F(X)$ for some smooth equivariant map X into some orthogonal representation. By (2) the realizable tensors form a convex cone.

2.2. **Hölder spaces and the group action.** Let ∇ be the Levi-Civita connection of g; it acts on V-valued tensor fields componentwise. For a V-valued tensor field T we use the Hölder norm

$$\|T\|_{k,\alpha}=\sum_{j=0}^{k}\sup_M|\nabla^jT|+[\nabla^kT]_\alpha,\qquad [S]_\alpha=\sup_{0<d(x,y)<\iota}\frac{|S(x)-\Pi_{y\to x}S(y)|}{d(x,y)^\alpha},$$

where d is the Riemannian distance, ι is the injectivity radius and $\Pi_{y\to x}$ is parallel transport along the unique minimizing geodesic. This norm is equivalent to the norm defined with charts and a partition of unity. Let $C^{k,\alpha}(M;V)$ and $C^{k,\alpha}(S^2T^*M)$ be the resulting Banach spaces.

The group G acts on V-valued maps and on tensors by

$$(\sigma \cdot v)(x) = \rho(\sigma)\, v(\sigma^{-1}x), \qquad \sigma \cdot h = (\sigma^{-1})^* h. \tag{3}$$

The fixed points of the first action are exactly the equivariant maps, and those of the second are the invariant tensors.

Lemma 2.1. *The actions* (3) *are linear and isometric for* $\|\cdot\|_{k,\alpha}$.

Proof. An isometry σ of (M, g) preserves ∇, the distance, minimizing geodesics, the injectivity radius and parallel transport, so $\nabla^j(\sigma \cdot T) = \sigma \cdot \nabla^j T$ and every term of the norm is preserved. The factor $\rho(\sigma)$ is orthogonal, so it preserves pointwise norms. □

For an arbitrary equivalent norm, $\|v\|' = \sup_{\sigma \in G} \|\sigma \cdot v\|$ is an equivalent norm for which G acts isometrically. So the choice of norm is not essential.

2.3. **Elliptic facts.** Let $\Delta = \operatorname{tr}_g \nabla^2$ be the rough Laplacian on scalar or V-valued tensor fields. It is nonpositive and self-adjoint, so $\Delta - 1$ is invertible on each tensor bundle. We write P_0 and P_1 for $(\Delta - 1)^{-1}$ on symmetric 2-tensors and on 1-forms. We use these standard facts.

- (E1) (Schauder estimates) P_0 and P_1 are bounded from $C^{k,\alpha}$ to $C^{k+2,\alpha}$ for every $k \geq 0$; see e.g. [3, Ch. 6] for the local estimates and [1] for compact manifolds.
- (E2) (Naturality) Δ, P_0, P_1 and Ric commute with isometries.
- (E3) (Products) Pointwise contractions define bounded bilinear maps $C^{k,\alpha} \times C^{k,\alpha} \to C^{k,\alpha}$, and multiplication by a smooth tensor is bounded on $C^{k,\alpha}$.
- (E4) (Commutators) For a V-valued function f, $\Delta \nabla f = \nabla \Delta f + \operatorname{Ric}(\nabla f)$. For a 1-form w, the commutator $\mathcal{K}(w) = \operatorname{Sym}([\Delta, \nabla]w)$ is a first-order operator whose coefficients are built from the curvature tensor and its covariant derivative. In particular $\mathcal{K}\colon C^{k+1,\alpha} \to C^{k,\alpha}$ is bounded and commutes with isometries.

3. Free maps and the pointwise right inverse

For a smooth map $u\colon M \to V$ and $x \in M$, define

$$A_u(x)\colon T_xM \oplus S^2T_xM \to V, \qquad A_u(x)(\xi, s) = du_x(\xi) + \nabla^2 u_x(s),$$

where the Hessian $\nabla^2 u_x \in S^2T_x^*M \otimes V$ is contracted with s.

Definition 3.1. The map u is *free* if $A_u(x)$ is injective for every $x \in M$.

In coordinates, $\partial_i\partial_j u = \nabla^2_{ij} u + \Gamma^k_{ij}\partial_k u$. So u is free if and only if the $n + \frac{1}{2}n(n+1)$ vectors $\partial_i u$ and $\partial_i\partial_j u$ $(i \leq j)$ are linearly independent. This is Nash's condition, called *perturbable* in [10].

Give $T_xM \oplus S^2T_xM$ the inner product induced by g and let $A_u(x)^\dagger$ be the adjoint. After identifying vectors and covectors with g,

$$A_u^\dagger y = (\langle du, y\rangle,\ \langle \nabla^2 u, y\rangle), \qquad \text{i.e. } (A_u^\dagger y)_i = \langle \partial_i u, y\rangle, \quad (A_u^\dagger y)_{ij} = \langle \nabla^2_{ij} u, y\rangle. \tag{4}$$

For free u define

$$L_u(x) = A_u(x)\big(A_u(x)^\dagger A_u(x)\big)^{-1}\colon T_x^*M \oplus S^2T_x^*M \to V.$$

Proposition 3.2 (Lean: `nash_gunther_right_inverse`)**.** *Let* $A\colon W \to V$ *be an injective linear map of finite-dimensional real inner product spaces and* $b \in W$*. Then* $x = A(A^\dagger A)^{-1}b$ *is the unique solution of* $A^\dagger x = b$ *in* $\operatorname{range} A$*, and it has the least norm among all solutions of* $A^\dagger x = b$*.*

Proof. $A^\dagger A$ is injective because $\langle A^\dagger A w, w\rangle = |Aw|^2$, hence invertible. If $x = Aw$ and $A^\dagger x = b$ then $w = (A^\dagger A)^{-1}b$. If $A^\dagger y = b$, then $y - x \in \ker A^\dagger = (\operatorname{range} A)^\perp$, so $|y|^2 = |x|^2 + |y - x|^2$. □

So $L_u(x)$ is a right inverse of $A_u(x)^\dagger$, depends smoothly on x, and does not depend on choices of coordinates. The next statement is the pointwise heart of equivariance in Step 1 of [10].

Proposition 3.3 (Lean: `nash_gunther_right_inverse_equivariant`)**.** *Let $A, A' \colon W \to V$ with A injective, let R be a linear isometry of V and P a linear automorphism of W, and suppose $A' = R \circ A \circ P$. If $x \in \operatorname{range} A$ solves $A^\dagger x = b$ and $x' \in \operatorname{range} A'$ solves $A'^\dagger x' = P^\dagger b$, then $x' = Rx$.*

Proof. $Rx \in \operatorname{range} A'$ since P is onto, and $A'^\dagger Rx = P^\dagger A^\dagger R^{-1} Rx = P^\dagger b$ since $R^\dagger = R^{-1}$. Apply the uniqueness in Proposition 3.2. □

Corollary 3.4. *Let u be a free equivariant map and b a G-invariant section of $T^*M \oplus S^2T^*M$ of class $C^{k,\alpha}$. Then $x \mapsto L_u(x)b(x)$ is an equivariant map of class $C^{k,\alpha}$.*

Proof. Differentiating $u \circ \sigma = \rho(\sigma) \circ u$ and using (E2) for ∇ gives $A_u(\sigma x) \circ Q = \rho(\sigma) \circ A_u(x)$, where $Q = d\sigma_x \oplus S^2 d\sigma_x$ is an isometry. Apply Proposition 3.3 with $A = A_u(x)$, $A' = A_u(\sigma x)$, $R = \rho(\sigma)$ and $P = Q^{-1}$. Then $P^\dagger = Q$ and $Qb(x) = b(\sigma x)$ by invariance, so $L_u(\sigma x) b(\sigma x) = \rho(\sigma) L_u(x) b(x)$. Regularity follows from (E3). □

4. A free equivariant starting embedding

By the theorem of Mostow and Palais [11, 13] there is an equivariant embedding $e \colon M \to W$ into an orthogonal representation W. Give $S^2W \subset W \otimes W$ the induced inner product, so that $S^2\rho$ is orthogonal, and put

$$u_0 = (e,\ e \odot e) \colon M \to W \oplus S^2W, \qquad e \odot e = e \otimes e.$$

This is the construction of [10, §2]; in an orthonormal basis it is $(x^a,\ (x^a)^2,\ \sqrt{2}\, x^a x^b)_{a<b}$.

Proposition 4.1 (Lean: `veronese_map_free`)**.** *u_0 is a free equivariant embedding.*

Proof. Equivariance and injectivity are clear. Fix a point, write $x = e(p)$, $x_i = \partial_i e(p)$ (linearly independent) and $x_{ij} = \partial_i \partial_j e(p)$, and write $a \odot b = a \otimes b + b \otimes a$. By the product rule,

$$\partial_i u_0 = (x_i,\ x_i \odot x), \qquad \partial_i \partial_j u_0 = (x_{ij},\ x_{ij} \odot x + x_i \odot x_j)$$

(for $\partial_i \partial_j$ of $e \otimes e$ the last term is $x_i \otimes x_j + x_j \otimes x_i$). Suppose $\sum c_i \partial_i u_0 + \sum d_{ij} \partial_i \partial_j u_0 = 0$ with d symmetric. The first component gives $a := \sum c_i x_i + \sum d_{ij} x_{ij} = 0$. Evaluating the second component on $\varphi \otimes \psi$ for linear functionals φ, ψ gives

$$\varphi(a)\psi(x) + \varphi(x)\psi(a) + 2 \sum_{i,j} d_{ij} \varphi(x_i) \psi(x_j) = 0.$$

Choosing φ, ψ dual to x_k, x_l gives $d_{kl} = 0$, and then $\sum c_i x_i = 0$ gives $c = 0$. □

Since tu_0 is free for $t \neq 0$, Proposition 4.1 supplies free equivariant embeddings with induced metric $t^2 F(u_0)$.

5. Günther's reformulation on a Riemannian manifold

Let u be a smooth free equivariant embedding into V. We look for $v \in C^{2,\alpha}(M; V)$ with $F(u + v) = F(u) + h$, where h is a given invariant symmetric 2-tensor. Write $(\operatorname{Sym} \nabla w)_{ij} = \frac{1}{2}(\nabla_i w_j + \nabla_j w_i)$ for a 1-form w.

5.1. **Two identities.** For V-valued u and v,

$$\langle \nabla_i u, \nabla_j v\rangle + \langle \nabla_j u, \nabla_i v\rangle = 2\,\mathrm{Sym}\,\nabla\big(\langle du, v\rangle\big)_{ij} - 2\langle \nabla^2_{ij} u, v\rangle, \tag{5}$$

since $\nabla_i\langle \nabla_j u, v\rangle = \langle \nabla^2_{ij} u, v\rangle + \langle \nabla_j u, \nabla_i v\rangle$ and the Hessian is symmetric.

The quadratic term $a_{ij} = \langle \nabla_i v, \nabla_j v\rangle$ is handled by Günther's trick. For V-valued $v, w \in C^{2,\alpha}$ define the 1-form and 2-tensor

$$\begin{aligned}\sigma(v,w)_j &= \tfrac{1}{2}\big(\langle \nabla_j v, \Delta w\rangle + \langle \nabla_j w, \Delta v\rangle\big),\\ \tilde\tau(v,w)_{ij} &= \mathrm{Sym}_{(ij),(vw)}\Big[-2\langle \nabla^2_{ij} v, \Delta w\rangle + 2\langle \nabla_i \nabla_k v, \nabla_j \nabla^k w\rangle\\ &\qquad + 2\,\mathrm{Ric}^k_i \langle \nabla_k v, \nabla_j w\rangle - \langle \nabla_i v, \nabla_j w\rangle\Big],\end{aligned}$$

where $\mathrm{Sym}_{(ij),(vw)}$ symmetrizes in i,j and in v,w. Both lie in $C^{0,\alpha}$ by (E3). Put

$$\mathcal{S}(v,w) = P_1\,\sigma(v,w), \qquad \mathcal{T}(v,w) = P_0\big(\tilde\tau(v,w) - 2\,\mathcal{K}(\mathcal{S}(v,w))\big). \tag{6}$$

By (E1) and (E4), $\mathcal{S}(v,w) \in C^{2,\alpha}(T^*M)$ and $\mathcal{T}(v,w) \in C^{2,\alpha}(S^2T^*M)$.

Lemma 5.1 (Günther's identity)**.** *For $v \in C^{2,\alpha}(M;V)$,*

$$\langle \nabla_i v, \nabla_j v\rangle = 2\,\mathrm{Sym}\,\nabla\big(\mathcal{S}(v,v)\big)_{ij} + \mathcal{T}(v,v)_{ij}.$$

Proof. First let v be smooth. By the Leibniz rule and (E4), applied to the V-valued 1-form ∇v,

$$\Delta a_{ij} = \langle \nabla_i \Delta v, \nabla_j v\rangle + \langle \nabla_i v, \nabla_j \Delta v\rangle + \mathrm{Ric}^k_i\langle \nabla_k v, \nabla_j v\rangle + \mathrm{Ric}^k_j\langle \nabla_k v, \nabla_i v\rangle + 2\langle \nabla_i\nabla_k v, \nabla_j\nabla^k v\rangle.$$

Using $\langle \nabla_i\Delta v, \nabla_j v\rangle = \nabla_i\langle \nabla_j v, \Delta v\rangle - \langle \nabla^2_{ij} v, \Delta v\rangle$, this gives $(\Delta - 1)a = 2\,\mathrm{Sym}\,\nabla\sigma(v,v) + \tilde\tau(v,v)$. Let $S = \mathcal{S}(v,v) = P_1\sigma(v,v)$. By the definition of $\mathcal{K}$, $(\Delta-1)\,\mathrm{Sym}\,\nabla S = \mathrm{Sym}\,\nabla\big((\Delta-1)S\big) + \mathcal{K}(S) = \mathrm{Sym}\,\nabla\sigma(v,v) + \mathcal{K}(S)$. Hence

$$(\Delta - 1)\big(a - 2\,\mathrm{Sym}\,\nabla S\big) = \tilde\tau(v,v) - 2\mathcal{K}(S),$$

and applying P_0 gives the claim. For general $v \in C^{2,\alpha}$, choose $\beta \in (0,\alpha)$. Smooth maps are dense in $C^{2,\alpha}$ for the $C^{2,\beta}$ norm, and both sides of the identity are continuous from $C^{2,\beta}$ to $C^{0,\beta}$. □

5.2. **The fixed point equation.** Combining (5) and Lemma 5.1,

$$F(u+v) - F(u) = 2\,\mathrm{Sym}\,\nabla\big(\langle du, v\rangle + \mathcal{S}(v,v)\big) - 2\Big(\langle \nabla^2 u, v\rangle - \tfrac{1}{2}\mathcal{T}(v,v)\Big).$$

So $F(u+v) = F(u) + h$ holds as soon as

$$\langle du, v\rangle = -\mathcal{S}(v,v), \qquad \langle \nabla^2 u, v\rangle = \tfrac{1}{2}\mathcal{T}(v,v) - \tfrac{1}{2}h. \tag{7}$$

By (4) this system is $A_u^\dagger v = \big(-\mathcal{S}(v,v),\ \tfrac{1}{2}\mathcal{T}(v,v) - \tfrac{1}{2}h\big)$. Define

$$\mathcal{M}(h) = L_u\big(0, -\tfrac{1}{2}h\big), \qquad \mathcal{B}(v,w) = L_u\big(-\mathcal{S}(v,w),\ \tfrac{1}{2}\mathcal{T}(v,w)\big),$$

where L_u acts pointwise.

Proposition 5.2. *Let $E = C^{2,\alpha}(M;V)$ and $\mathrm{Met}^{2,\alpha} = C^{2,\alpha}(S^2T^*M)$.*

(a) $\mathcal{M}\colon \mathrm{Met}^{2,\alpha} \to E$ is bounded linear and $\mathcal{B}\colon E\times E \to E$ is bounded bilinear.
(b) If $v \in E$ satisfies $v = \mathcal{M}(h) + \mathcal{B}(v,v)$, then $F(u+v) = F(u) + h$.
(c) If h is G-invariant then $\mathcal{M}(h)$ is G-fixed, and $\mathcal{B}(\sigma\cdot v, \sigma\cdot w) = \sigma\cdot\mathcal{B}(v,w)$ for all $\sigma \in G$.

Proof. (a) follows from (E1)–(E4), because L_u has smooth coefficients. For (b), $A_u^\dagger L_u = \mathrm{id}$ gives $A_u^\dagger v = (0, -\frac{1}{2}h) + (-\mathcal{S}(v,v), \frac{1}{2}\mathcal{T}(v,v))$, which is (7). For (c), the operators ∇, Δ, Ric, P_0, P_1 and $\mathcal{K}$ and the contractions with g are natural under isometries (E2), and the inner product of V is ρ-invariant. So $\mathcal{S}(\sigma\cdot v, \sigma\cdot w) = \sigma\cdot\mathcal{S}(v,w)$, and likewise for $\mathcal{T}$. Corollary 3.4 (in its pointwise form) gives the equivariance of L_u. □

We also need two standard facts.

(F1) (Regularity, Günther [4, 5]) There is $r > 0$ such that if h is smooth, $v \in E$, $\|v\|_{2,\alpha} < r$ and $v = \mathcal{M}(h) + \mathcal{B}(v,v)$, then v is smooth. The reason is that, after differentiating the equation k times, the top-order derivatives of v satisfy a *linear* equation $w = f + \mathcal{L}_v w$, where $\|\mathcal{L}_v\| \le C\|v\|_{2,\alpha}$ and C does not depend on k.

(F2) (Openness of embeddings) There is $\delta > 0$ such that if $\|v\|_{2,\alpha} < \delta$ then $u + v$ is an embedding; see [6, Ch. 2].

6. The equivariant Günther perturbation theorem

Theorem 6.1 (Lean: `gunther_fixed_point_equivariant`)**.** *Let E be a real Banach space on which a monoid G acts by linear isometries. Let $\mathcal{B}\colon E\times E\to E$ be bounded bilinear and G-equivariant, and let $m \in E$ be G-fixed with $4\|\mathcal{B}\|\,\|m\| < 1$. Then $v = m + \mathcal{B}(v,v)$ has a unique solution with $\|v\| \le 2\|m\|$, and this solution is G-fixed.*

Proof. Let $T(v) = m + \mathcal{B}(v,v)$ and $\bar{B} = \{\|v\| \le 2\|m\|\}$. For $v, w \in \bar{B}$ we have $\|T(v)\| \le \|m\| + 4\|\mathcal{B}\|\|m\|^2 \le 2\|m\|$, and $\|T(v) - T(w)\| = \|\mathcal{B}(v, v-w) + \mathcal{B}(v-w, w)\| \le 4\|\mathcal{B}\|\|m\|\,\|v-w\|$. So T is a contraction of the complete set $\bar{B}$ and has a unique fixed point v there. For $\sigma \in G$, the point $\sigma\cdot v$ lies in $\bar{B}$ (the action is isometric) and $T(\sigma\cdot v) = \sigma\cdot m + \sigma\cdot\mathcal{B}(v,v) = \sigma\cdot T(v) = \sigma\cdot v$. By uniqueness, $\sigma\cdot v = v$. □

Lean proves this for a general quadratic map Q with $Q(0) = 0$ and $\|Q(v) - Q(w)\| \le C(\|v\| + \|w\|)\|v - w\|$, and deduces the bilinear case.

Theorem 6.2 (Equivariant perturbation; Lean: `equivariant_gunther_perturbation`)**.** *Let u be a smooth free equivariant embedding. There is $\varepsilon > 0$ such that for every smooth G-invariant symmetric 2-tensor h with $\|h\|_{2,\alpha} < \varepsilon$ there is a smooth equivariant embedding X (for the same representation) with $F(X) = F(u) + h$.*

Proof. Use the equivariant Banach space $E = C^{2,\alpha}(M; V)$ (Lemma 2.1) and the operators of Proposition 5.2. Let r, δ be as in (F1) and (F2), let

$$\eta = \min\Big(\tfrac{1}{2}\min(\delta, r),\ \frac{1}{4\|\mathcal{B}\| + 1}\Big), \qquad \varepsilon = \frac{\eta}{\|\mathcal{M}\| + 1}.$$

If $\|h\|_{2,\alpha} < \varepsilon$ then $m = \mathcal{M}(h)$ is G-fixed with $\|m\| < \eta$, so $4\|\mathcal{B}\|\|m\| < 1$. Theorem 6.1 gives a G-fixed v with $v = m + \mathcal{B}(v,v)$ and $\|v\| \le 2\|m\| < \min(\delta, r)$. By (F1), v is smooth; it is equivariant since it is G-fixed; $X = u + v$ is an embedding by (F2); and $F(X) = F(u) + h$ by Proposition 5.2(b). □

Remark 6.3. Theorem 6.2 needs no averaging of solutions. Averaging is in fact useless here: the average of two solutions of $F(X) = g$ is not a solution. Equivariance comes entirely from the uniqueness of the small fixed point. Equivalently, the contraction preserves the closed subspace E^G of equivariant maps, and one may run the iteration inside E^G.

7. Approximation by realizable metrics

This step is that of [10, §3]; we include it for completeness and to point out the part that is formalized.

Proposition 7.1. *Every smooth positive definite G-invariant metric g' on M is a limit, in every $C^{k,\alpha}$ norm, of realizable metrics.*

Proof. By the (non-equivariant) Nash embedding theorem, for instance in Günther's form, there are smooth functions $y^1, \dots, y^q$ with $g' = \sum_i dy^i \otimes dy^i$. (The weaker statement that g' can be approximated by such sums, proved directly in [12, p. 58], is enough.) Let Δ_g be the Laplace–Beltrami operator of g. The partial sums of the eigenfunction expansion of a smooth function converge in every Sobolev norm, hence in every C^k norm. So there are finite sums f_i of eigenfunctions with

$$\Big\| g' - \sum_i df_i \otimes df_i \Big\|_{k,\alpha} < \varepsilon.$$

Each eigenspace of Δ_g is finite-dimensional and G-invariant (G acts by isometries). So the span V_i of the translates $\sigma \cdot f_i$ is a finite-dimensional G-invariant space of functions. Let $X_i \colon M \to V_i^*$ be the evaluation map $X_i(p)(f) = f(p)$; it is smooth and equivariant for the dual representation. Let $\varphi_i \in V_i^{**}$ be evaluation at f_i, and give V_i^* the inner product

$$\langle \xi, \eta \rangle_i = \int_G \xi(\sigma \cdot f_i)\, \eta(\sigma \cdot f_i)\, d\sigma = \int_G \varphi_i(\sigma^{-1}\xi)\, \varphi_i(\sigma^{-1}\eta)\, d\sigma, \tag{8}$$

with $d\sigma$ normalized Haar measure. By Proposition 7.2 below, this is a G-invariant inner product, since the translates of f_i span V_i. For tangent vectors a, b at p,

$$F(X_i)(a,b) = \int_G d(\sigma \cdot f_i)(a)\, d(\sigma \cdot f_i)(b)\, d\sigma = \Pi(df_i \otimes df_i)(a,b), \qquad \Pi(\kappa) = \int_G \sigma^* \kappa \, d\sigma,$$

where the Haar projection Π is computed pointwise. Since Π commutes with ∇ and with parallel transport, Lemma 2.1 gives $\|\Pi\kappa\|_{k,\alpha} \le \|\kappa\|_{k,\alpha}$ (estimate each term of the norm inside the integral). Also $\Pi g' = g'$. So $X = (X_1, \dots, X_q)$ is a smooth equivariant map with

$$\|g' - F(X)\|_{k,\alpha} = \Big\| \Pi\Big(g' - \sum_i df_i \otimes df_i \Big) \Big\|_{k,\alpha} < \varepsilon. \qquad \square$$

Proposition 7.2 (Lean: `haar_averaged_inner_product`)**.** *Let a compact group G act linearly and continuously on a finite-dimensional real vector space W. Let μ be a finite left-invariant measure on G that is positive on nonempty open sets, and let $\varphi \in W^*$ be such that the functionals $\varphi \circ \rho(\sigma)$ separate the points of W. Then $q(x,y) = \int_G \varphi(\rho(\sigma)^{-1}x)\, \varphi(\rho(\sigma)^{-1}y)\, d\mu(\sigma)$ is a G-invariant inner product on W.*

Proof. Bilinearity and symmetry are clear, and invariance follows from left-invariance of μ. If $x \neq 0$, the continuous function $\sigma \mapsto \varphi(\rho(\sigma)^{-1}x)^2$ is nonnegative and not identically zero, so its integral is positive. $\square$

Remark 7.3. The construction can be made explicit. Decompose $L^2(M)$ (complex-valued) into irreducible unitary representations of G inside the eigenspaces, expand the approximating functions z^i in bases of these, and regroup the coefficients by isomorphism type. This gives an equivariant map X with $\sum_{a,r} N_r^{1/2} X_{iaar} = z^i$ and $F(X) = \Pi(\sum_i dz^i \otimes dz^i)$; see [15].

8. Proof of the Main Theorem

Let g be the given invariant metric. Proposition 4.1 gives a free equivariant embedding u_0. Since M is compact there is $c > 0$ with $g - cF(u_0)$ positive definite. The map $u_1 = \sqrt{c}\, u_0$ is a free equivariant embedding with $F(u_1) = cF(u_0)$. Let ε be given by Theorem 6.2 for u_1. By Proposition 7.1 there is a smooth equivariant map X_2 with

$$\|h\|_{2,\alpha} < \varepsilon, \qquad h := g - cF(u_0) - F(X_2).$$

The tensor h is smooth and G-invariant. Theorem 6.2 gives a smooth equivariant embedding X_1 with $F(X_1) = F(u_1) + h = g - F(X_2)$. Then $X = (X_1, X_2)$ is an equivariant embedding into the orthogonal representation $V_1 \oplus V_2$, and $F(X) = F(X_1) + F(X_2) = g$ by (2). □

This argument is formalized in Lean as `equivariant_isometric_embedding_conditional`, with the inputs listed in Table 1 as hypotheses.

Remark 8.1. As in [10], the case of manifolds with boundary reduces to the closed case: embed M equivariantly in a sphere in an orthogonal representation and extend the invariant metric to an invariant metric on the sphere by an invariant partition of unity. We do not treat the real-analytic category, where [10, §4] uses the method of Greene and Jacobowitz.

9. The Lean formalization

The formalization is available at
https://github.com/schlafly1/equivariant-embedding.
It uses Lean 4 (toolchain `v4.35.0`) and Mathlib (tag `v4.35.0`), and follows the layout of the Palomar registry: a short file `Challenge.lean` with the advertised statements (proofs replaced by `sorry`), a file `Solution.lean` with the same statements proved, and the proof development in the directory `EquivEmbed/`. Every theorem in `Solution.lean` depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`; `Lean.ofReduceBool` (`native_decide`) is not used.

9.1. Unconditional statements.

Lean declaration	Mathematical content	Here
`gunther_fixed_point_equivariant`	equivariant Günther fixed point	Thm. 6.1
`nash_gunther_right_inverse`	pointwise right inverse, least norm	Prop. 3.2
`nash_gunther_right_inverse_equivariant`	equivariance, coordinates	Prop. 3.3
`veronese_map_free`	freeness of $(e, e \odot e)$	Prop. 4.1
`haar_averaged_inner_product`	averaged invariant inner product	Prop. 7.2

These are stated with ordinary Mathlib notions: Banach spaces with an isometric action (`IsIsometricSMul`), adjoints of linear maps between finite-dimensional inner product spaces, linear functionals, Bochner integrals with respect to a left-invariant measure, and `Representation`.

9.2. **The conditional main theorem.** Maps are elements of the structure `EquivariantMap`: a function $X\colon \texttt{Pt} \to \mathbb{R}^N$ on a type `Pt` with a G-action, bundled with a homomorphism ρ from G to the linear isometries of $\mathbb{R}^N$ and a proof that $X(\sigma p) = \rho(\sigma)X(p)$. Equivariance is thus enforced by the types, and the conclusions of the theorems restate it. An abstract structure `EmbeddingProblem` records the type `Pt` (the points of M), a normed space `Met` (intended: invariant $C^{2,\alpha}$ tensors), a subset `pos` (intended: positive definite tensors), the induced metric F from equivariant maps to `Met`, predicates for being an embedding (required to imply injectivity) and being free, and the direct sum, with the identities (2) for the sum and for scaling. The

theorem `equivariant_isometric_embedding_conditional` states that every element of `pos` equals $F(X)$ for an equivariant embedding X, under the four hypotheses in Table 1.

The geometric meaning of `Met`, `pos`, F, freeness and being an embedding (beyond injectivity) is not encoded in Lean. The theorem holds for every instantiation of `EmbeddingProblem`; its reading as a statement about Riemannian manifolds depends on an instantiation for an actual compact Riemannian G-manifold, which is given by Sections 3–8 but is not formalized.

TABLE 1. Hypotheses of the conditional main theorem.

Lean hypothesis	Content	Justification
`HasFreeEmbedding`	a free equivariant embedding exists	Mostow–Palais and Prop. 4.1
`RealizableDense`	realizable metrics are dense in `pos`	Prop. 7.1
`ConeCondition`	$g - cF(X) \in$ `pos` for some $c > 0$ (one c suffices)	compactness of M
`HasGuntherPackages`	at each free embedding: E, $\mathcal{M}$, $\mathcal{B}$ as in Prop. 5.2, (F1), (F2)	Sections 3–5

The structure `GuntherPackage` asks for a Banach space E with an isometric linear G-action, a bounded linear $\mathcal{M}$ with G-fixed values, a bounded bilinear equivariant $\mathcal{B}$, a map sending smooth G-fixed v to a map "$u+v$", Günther's identity (Proposition 5.2(b)), regularity of small solutions, and openness of embeddings. These are non-equivariant analytic facts together with the naturality of the operators; the existence of an equivariant solution is not assumed. It is produced in Lean by Theorem 6.1, through `equivariant_gunther_perturbation`.

9.3. **What is not formalized.** The objects of differential geometry (manifolds, Hölder spaces of sections, the Laplacian, Schauder estimates, eigenfunction expansions, the Mostow–Palais theorem, the openness of embeddings) are not formalized; they enter only through the hypotheses above. The link between the abstract structure and an actual compact Riemannian G-manifold is therefore given by the paper, not by Lean. Formalizing it would need substantial new Mathlib infrastructure, in particular elliptic regularity on compact manifolds.

9.4. **Checking.** The project builds with `lake build`. The script `scripts/gen_challenge.py` generates `Challenge.lean` and `Solution.lean` from a single list of statements, and copies the definitions of `EquivEmbed/Defs.lean` verbatim into `Challenge.lean`. We checked that the fully elaborated types of all compared declarations agree in the two environments, and that `#print axioms` reports only the three standard axioms. The continuous integration configuration runs Lean's `lake comparator` with the independent kernels shipped with the toolchain.

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